

XIN LIU (刘鑫)

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EDUCATION

National University of Singapore (NUS)

Singapore

Visiting PhD in Electrical and Computer Engineering (Advised by Prof. Fan Shi)

2025.08 – now

- Architecting simulator-in-the-loop frameworks for manipulation capability on single- and dual-arm robots.
- Implementing world action models (WAM) for robotic manipulator control.

Shanghai Jiao Tong University (SJTU)

Shanghai, China

PhD in Mechanical Engineering (Advised by Prof. Chenkun Qi)

2023.09 – now

- Developed RL and IL driven whole-body loco-manipulation policies, enabling legged robots to perform complex tasks while maintaining robust stability.

M.S. in Mechanical Engineering

2021.09 - 2023.09

- Developed robust locomotion DRL policies for legged robots, focusing on gait generation and terrain adaptability.
- Developed adaptive SMC with parameter identification for high-precision 6-DOF parallel robot tracking.

Xi'an Jiaotong University (XJTU)

Xi'an, China

B.E. in Mechanical Engineering

B.E. in Computer Science and Technology

2017.09 - 2021.07

Grade: 3.91/4.3 (Rank: 3/212)

- Bachelor's Thesis: A Spherical Robot for Landing and Barrier Breaking (Major); Target Recognition, Detection, Segmentation and Camera Calibration in Robot Grabbing Tasks (Minor).
- Core Coursework: Mechanical Control System Engineering, Advanced Programming, Microcomputer principle and Interface technology, Data Structures and Algorithms, Computer Organization, Introduction to Internet of Things.

INTERNSHIP

Gradient Network (INFINILINK PTE. LTD.)

Singapore

Robotics Simulation Intern

2025.10 – 2026.03

- Developed a visual-based DRL algorithm on a proprietary differentiable simulator to enable complex deformable object manipulation, achieving sim-to-real transfer on a dual-arm humanoid robot and a desktop robotic manipulator.

Shanghai Artificial Intelligence Laboratory (Shanghai AI Lab)

Shanghai, China

Embodied AI Intern (Advised by Yan Ding)

2024.10 – 2025.06

- Developed MLM, a unified RL framework for whole-body loco-manipulation on a quadruped robot, enabling coordinated movement and interaction in dynamic environments.
- Developed Fast-UMI, a hardware-independent manipulation interface and large-scale dataset designed to generalize Imitation Learning strategies across diverse robotic platforms.

Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences (CSU)

Beijing, China

Research Intern

2019.07 - 2019.08

- Worked on single-sided two-way ranging technology and double-sided two-way ranging technology.

PUBLICATIONS

- Enabling Robust Cloth Manipulation via Inference-Time Simulator-in-the-Loop Refinement, 2026. [website](#), [paper](#)
X. Liu, Y. Li, Z. Li, P. Jing, Z. Huang, B. Zhou, Z. Zeng, S. Luo, C. Qi, F. Shi
- **FLASH**: Fast Learning via GPU-Accelerated Simulation for High-Fidelity Deformable Manipulation in Minutes, 2026. [paper](#)
S. Luo, B. Zhou, C. Zhang, **X. Liu**, Z. Huang, G. Yang, Z. Han, X. Hu, E. Yang, R. Yu, Z. Zeng, F. Shi
- **MLM**: Learning Multi-task Loco-Manipulation Whole-Body Control for Quadruped Robot with Arm, IEEE Robotics and Automation Letters (**RAL**), 2026. [video](#), [paper](#)
X. Liu, B. Ma, C. Qi, Y. Ding et al.
- Skill latent space based multi-gait learning for a legged robot, IEEE Transactions on Industrial Electronics (**TIE**), 2025. [video](#), [paper](#)
X. Liu*, J. Z. Wu*, Y. F. Xue, C. K. Qi, G. Y. Xin, and F. Gao
- Learning natural and robust hexapod locomotion over complex terrains via motion priors based on deep reinforcement learning, (**IROS 2025**). [video](#), [paper](#)
X. Liu, J. Z. Wu, Y. H. Li, C. K. Qi, Y. F. Xue, and F. Gao
- **FastUMI**: A Scalable and Hardware-Independent Universal Manipulation Interface with Dataset, (**CoRL 2025**). [website](#), [paper](#)
Zhaxizhuoma*, K. Liu*, C. Guan*, Z. Jia*, Z. Wu*, **X. Liu***, T. Wang, S. Liang, P. Chen et al.
- **FastUMI-100K**: Advancing Data-driven Robotic Manipulation with a Large-scale UMI-style Dataset, 2025.
K. Liu, Z. Jia, Y. Li, P. Chen, S. Liu, **X. Liu**, P. Zhang, H. Song et al.
- **PPL**: Point Cloud Supervised Proprioceptive Locomotion Reinforcement Learning for Legged Robots in Crawl Spaces, IEEE Robotics and Automation Letters (**RAL**), 2026. [video](#), [paper](#)
B. Ma, N. Xu, C. Qi, **X. Liu**, Y. Mo, J. Wang, and C. Lu
- Experience-Learning Inspired Two-Step Reward Method for Efficient Legged Locomotion Learning towards Natural and Robust Gaits, (**IROS 2024**). [paper](#)
Y. H. Li, J. Z. Wu, **X. Liu**, W. Z. Guo, and Y. F. Xue
- An actuation acceleration based kinematic modeling and parameter identification approach for a six-DOF 6-PSU parallel robot with joint clearances, ASME. J. Mechanisms Robotics (**ASME JMR**), 2025.
X. Liu, C. K. Qi, J. F. Lin, D. J. Li, and F. Gao
- Learning Graphic Convolutional Network Based Quadruped Locomotion on Complex Terrain, (**CAC 2024**).
X. Li, A. Li, **X. Liu**, Y. Xue, X. Xu, C. Qi
- Multi-input multi-output sliding mode control with high precision and robustness for a 6-PSU parallel robot, (**ICIRA 2023**).
X. Liu, J. F. Lin, D. J. Li, C. K. Qi, and F. Gao
- Modeling and identification of kinematic model with joint clearance for a 6-PSU parallel manipulator, (**CCC 2023**).
X. Liu, C. K. Qi, J. F. Lin, D. J. Li, and F. Gao
- A Motion-Adaptive Parallax-Based Monocular Visual Odometry, (**CCC 2023**).
Z. Xu, **X. Liu**, C. Qi

PROFESSIONAL SERVICES

Reviewer of TRO, TIE, RA-L, CoRL, ICRA, IROS.

ACHIEVEMENTS

Shanghai Jiao Tong University

- Merit Student 2023, 2022
- The First Prize Scholarship 2021

Xi'an Jiaotong University

- Outstanding Thesis for Bachelor's Degree (Top 1%) 2021
- Outstanding Graduate 2021
- HIWIN Outstanding Student Scholarship (Top 0.1%) 2020
- First Prize in the 9th National University Student Mechanical Innovation Design Competition 2020
- Merit Student 2020, 2018
- The First Prize Scholarship 2019, 2018

SKILLS AND INTERESTS

Digital Skills

Python | PyTorch | Isaac | MuJoCo | ROS | C/C++ | Linux | Git | Matlab

Language Skills

Other language: English (IELTS: 7.0)

Interests

Swimming, Tennis, Film, Football

刘鑫

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教育经历

新加坡国立大学 (NUS)

新加坡

电子与计算机工程 访问博士研究生 (导师：石凡 Fan Shi)

2025.08 – 至今

- 构建单臂和双臂机器人柔性布料操作能力的仿真在环 (simulator-in-the-loop) 框架，真机部署。
- 研究机器人柔性布料操作能力的世界动作模型 (WAM)。

上海交通大学 (SJTU)

上海

机械工程 博士研究生 (导师：齐臣坤)

2023.09 – 至今

- 开发基于强化学习 (RL) 和模仿学习 (IL) 的全身移动操作 (loco-manipulation) 策略，使足式机器人能够在保持稳定性的同时执行复杂任务。

机械工程 硕士研究生

2021.09 - 2023.09

- 开发足式机器人鲁棒的深度强化学习 (DRL) 运动策略，重点关注步态生成与地形适应能力。
- 针对高精度六自由度并联机器人轨迹跟踪，开发了结合参数辨识的自适应滑模控制 (SMC) 方法。

西安交通大学 (XJTU)

西安

机械工程 学士

计算机科学与技术 学士

2017.09 - 2021.07

GPA: 3.91/4.3 (Rank: 3/212)

- 学士学位论文：面向登陆破障的球形机器人（主修）；机器人抓取任务中的目标识别、检测、分割及相机标定（辅修）。
- 核心课程：机械控制工程、高级程序设计、微机原理与接口技术、数据结构与算法、计算机组成原理。

实习经历

Gradient Network (INFINILINK PTE. LTD.)

新加坡

机器人仿真实习生

2025.10 – 2026.03

- 基于自研可微仿真器开发一种视觉驱动的深度强化学习 (DRL) 算法，实现对复杂可变形柔性物体的操作，并在双臂人形机器人及桌面机械臂上成功实现了从仿真到现实 (sim-to-real) 的迁移。

上海人工智能实验室 (Shanghai AI Lab)

上海

具身 AI 实习生 (导师：丁琰)

2024.10 – 2025.06

- 开发了 MLM：一种用于四足机器人全身移动操作 (loco-manipulation) 的统一强化学习框架，实现了动态环境下的四足机器人移动操作能力。
- 开发了 Fast-UMI：一个独立于硬件的操作接口及大规模数据集，实现模仿学习策略在多种机器人平台间的泛化。

- 单向-双向测距技术及双向-双向测距技术的研究工作。

论文发表

- Enabling Robust Cloth Manipulation via Inference-Time Simulator-in-the-Loop Refinement, 2026. [website](#), [paper](#)
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- A Motion-Adaptive Parallax-Based Monocular Visual Odometry, (**CCC 2023**).
Z. Xu, **X. Liu**, C. Qi

学术服务

TRO, TIE, RA-L, CoRL, ICRA, IROS 等具身智能和机器人领域顶刊顶会审稿人。

成果

上海交通大学

- 三好学生2023, 2022
- 一等奖学金2021

西安交通大学

- 优秀毕业设计（前 1%）2021
- 优秀毕业生2021
- HIWIN 菁英学生奖学金（前 0.1%）2020
- 第九届全国大学生机械设计创新大赛 全国一等奖2020
- 三好学生2020, 2018
- 一等奖学金2019, 2018

技能和兴趣

专业技能

Python | PyTorch | Isaac | MuJoCo | ROS | C/C++ | Linux | Git | Matlab

语言技能

其他语言：英语（雅思：7.0）

兴趣

游泳，网球，电影，足球